

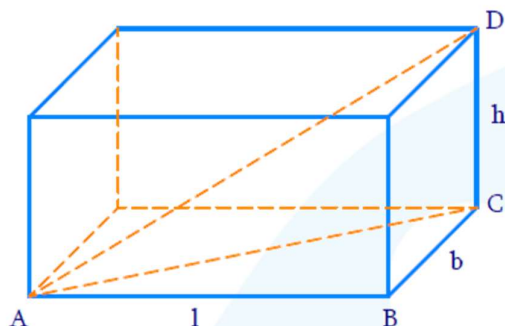
Mensuration and Solids

Review Exercise Answers

Level-1

Only Single Option Correct

S1. (B). Consider the following figure:



The length of the diagonal of this cuboid can be calculated as follows:

$$\begin{aligned} AD &= \sqrt{AC^2 + CD^2} \\ &= \sqrt{(AB^2 + BC^2) + CD^2} \\ &= \sqrt{l^2 + b^2 + h^2} \end{aligned}$$

Thus, the length of the longest pole will be the length of the diagonal, which will be $\sqrt{10^2 + 10^2 + 5^2} = 15$ m.

S2. (D). The required volume is

$$\begin{aligned} V &= \text{Base} \times \text{Height} \\ &= (120 \text{ cm}^2) \times (50 \text{ cm}) \\ &= 6000 \text{ cm}^3 = 6 \text{ L} \end{aligned}$$

S3. (A). Let the height be h m. We have:

$$\begin{aligned} \text{Volume} \times \text{Density} &= \text{Mass} \\ \Rightarrow (\pi r^2 h) \times d &= m \\ \Rightarrow \pi (0.8)^2 h \times (60) &= 100 \\ \Rightarrow h &\approx 0.83 \text{ m} = 83 \text{ cm} \end{aligned}$$

S4. (D). Let the side lengths of the two cubes be l_1 and l_2 . Using the given value for the ratio of the volumes, we have:

$$\frac{l_1^3}{l_2^3} = k \Rightarrow \frac{l_1}{l_2} = \sqrt[3]{k}$$

The ratio of the surface areas will be

$$\frac{6l_1^2}{6l_2^2} = \left(\sqrt[3]{k}\right)^2 = \sqrt[3]{k^2}$$

S5. (A). The surface area of a sphere varies as the square of the radius, while the volume varies as the cube of the radius. Thus, only the surface area will increase by a factor of 4.

S6. (B). The required volume is

$$\begin{aligned} & \frac{2}{3}\pi\{r^3 - (r-1)^3\} \\ &= \frac{2}{3}\pi(3r^2 - 3r + 1)\text{cm}^3 \end{aligned}$$

S7. (C). The height of the cone is $H = 2R$. Now, the radius and height of the top (conical) half of the cone is $\frac{R}{2}$ and $\frac{H}{2} = R$. We have:

$$V_{\text{entire cone}} = \frac{1}{3}\pi R^2 H = \frac{1}{3}\pi R^2 (2R) = \frac{2}{3}\pi R^3$$

$$V_{\text{top conical half}} = \frac{1}{3}\pi \left(\frac{R}{2}\right)^2 \left(\frac{H}{2}\right) = \frac{1}{12}\pi R^3$$

$$\begin{aligned} \Rightarrow V_{\text{bottom half}} &= V_{\text{entire cone}} - V_{\text{top conical half}} \\ &= \frac{2}{3}\pi R^3 - \frac{1}{12}\pi R^3 = \frac{7}{12}\pi R^3 \end{aligned}$$

The required ratio is

$$\frac{V_{\text{bottom half}}}{V_{\text{top conical half}}} = \frac{\frac{7}{12}\pi R^3}{\frac{1}{12}\pi R^3} = 7$$

Integer Answers

S8. We have:

$$4\pi r^2 = 314$$

$$\Rightarrow 4(3.14)r^2 = 314$$

$$\Rightarrow r^2 = 25 \Rightarrow r = 5$$

The radius of the sphere is 5 cm.

S9. The answer will be the ratio of the entire volume of the rod to the volume of an individual pellet:

$$n = \frac{V_{rod}}{V_{pellet}} = \frac{\pi(10)^2(30)}{\pi(1)^2(3)} = 1000$$

Thus, 1000 pellets will be formed.

S10. Let the breadth and height of the cuboid be $3x$ m and $2x$ m. We have:

$$V = lbh$$

$$\Rightarrow 1536 = (16)(3x)(2x)$$

$$\Rightarrow x = 4$$

The breadth and height are 12 m and 8 m respectively. The total surface area will be

$$\begin{aligned} & 2(lb + bh + lh) \\ &= 2(16 \times 12 + 12 \times 8 + 16 \times 8) \\ &= 2(192 + 96 + 128) \\ &= 832 \text{ m}^2 \end{aligned}$$

S11. The volume of the beam is

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(0.5)^2(9) \approx 2.355 \text{ m}^3$$

Thus, the mass of the beam is

$$(2.355 \text{ m}^3) \times (50 \text{ kg/m}^3) = 117.75 \text{ kg}$$

To the nearest integer, we can write the answer as 118 kg.

S12. Since $V = \frac{1}{3}\pi r^2 h$ for a cone, doubling the height doubles the volume, while tripling the radius increases the volume by a factor of $3^2 = 9$. Thus, the overall increase in volume will be by a factor of 18.

_____.

S13. The TSA of the original cube is $6(4)^2 = 96 \text{ cm}^2$. The total of the TSAs of the 64 smaller cubes is

$$64\{6(1)^2\} = 384 \text{ cm}^2$$

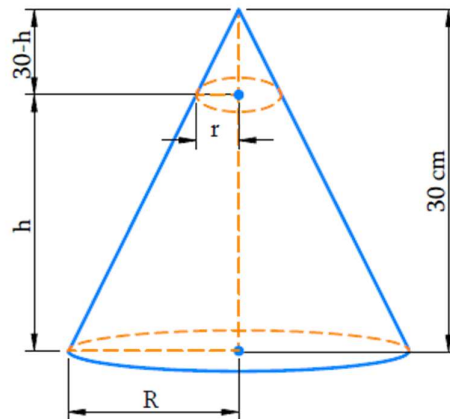
The increase in surface area is by a factor of

$$\frac{384 \text{ cm}^2}{96 \text{ cm}^2} = 4$$

S14. We divide the volume of the tank by that of the bucket:

$$n = \frac{(250) \times (150) \times (120)}{\pi \times (40)^2 \times (60)} \approx 14.9$$

S15. Consider the following figure:



We have assumed the radius of the complete cone to be R , while the radius of the small cut-off cone is r . We note that:

$$\frac{r}{R} = \frac{30-h}{30} = 1 - \frac{h}{30}$$

Now, the ratio of the volumes of the small cone and the big cone is $1/27$. We have:

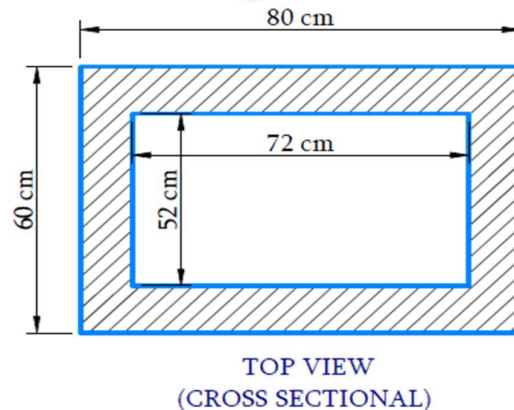
$$\begin{aligned} \frac{\frac{1}{3}\pi r^2(30-h)}{\frac{1}{3}\pi R^2(30)} &= \frac{1}{27} \\ \Rightarrow \left(\frac{r}{R}\right)^3 &= \frac{1}{27} \text{ (how?)} \\ \Rightarrow \frac{r}{R} &= \frac{1}{3} \end{aligned}$$

Thus,

$$1 - \frac{h}{30} = \frac{1}{3} \Rightarrow h = 20$$

The section is made 20 cm above the base.

S16. The outside dimensions of the base are $80 \text{ cm} \times 60 \text{ cm}$. Thus, the base dimensions of the cuboidal hollow inside the container are $72 \text{ cm} \times 52 \text{ cm}$. This is because we have to subtract $(4 \text{ cm} + 4 \text{ cm})$ along both the length and the breadth, due to the two vertical walls. The following figure makes this clear:



The outside height of the box is 40 cm, and so the inside height of the cuboidal hollow will be 40 cm – (12 cm + 3 cm), or 25 cm. Thus, the dimensions of the cuboidal hollow are 72 cm × 52 cm × 25 cm.

Observe the bar dimensions (18 cm × 13 cm × 5 cm):

- the length of the interior hollow is an integer multiple of the bar length
- similarly, the breadth and height of the hollow are integer multiples of the breadth and height of the bar

Clearly, we can arrange the bars inside the hollow so that the volume of the hollow is *completely* occupied. Thus, the number of bars which can be fit in the hollow is simply the ratio of the volume of the hollow to that of one bar:

$$n = \frac{72 \times 52 \times 25}{18 \times 13 \times 5} = 80$$

Remark. Many students, when asked this question, directly calculate this ratio, without verifying whether a *complete volume fit* is possible or not. This would be a mistake in a general scenario.

S17. Self-exercise. The answer can be obtained as the ratio of the volume of the bar to the volume of iron used to make one spherical shell. The answer is 161.

Miscellaneous

S18. Self-exercise. The answer is 1: 2.

S19. (i) The volume increases by a factor of 8. (ii) The volume increases by a factor of 27. (iii) The volume decreases by a factor of 8.

S20. (a) One hectare is 10,000 m². The total volume of water will be

$$\begin{aligned} V &= (2 \text{ cm}) \times (5 \text{ hectares}) \\ &= (0.02 \text{ m}) \times (50,000 \text{ m}^2) \\ &= 1000 \text{ m}^3 \end{aligned}$$

(b) Self-exercise. The answer is 3 cm of rainfall.

S21. (a) Let the original dimensions of the cuboid be $l \times b \times h$. The new dimensions will be $1.5l \times 1.5b \times 1.5h$. The new surface area will be:

$$\begin{aligned} &2(1.5l \times 1.5b + 1.5b \times 1.5h + 1.5l \times 1.5h) \\ &= 2.25 \times \{2(lb + bh + lh)\} \end{aligned}$$

Clearly, the new surface area is 2.25 times that of the original surface area, or 225 cm².

(b) The volume will increase by a factor of 1.728. The new volume will be 1.728 L, or 1728 cm³.

(c) Let the original dimensions of the cuboid be $l \times b \times h$. The new dimensions will be $1.2l \times 0.8b \times 1.2h$. The new volume is:

$$1.2l \times 0.8b \times 1.2h = 1.152lbh$$

Clearly, the new volume is 1.152 times the original volume – 1.152 L or (1152 cm³).

S22. Let B's side be x units and let the paint to be used on B cost INR y / unit². Then, the corresponding values for A will be $1.5x$ units and INR $0.8y$ / unit². The costs of the paint jobs will be:

$$C_A = 6(1.5x)^2 \times (0.8y) = 10.8x^2y$$

$$C_B = 6(x)^2 \times (y) = 6x^2y$$

Thus,

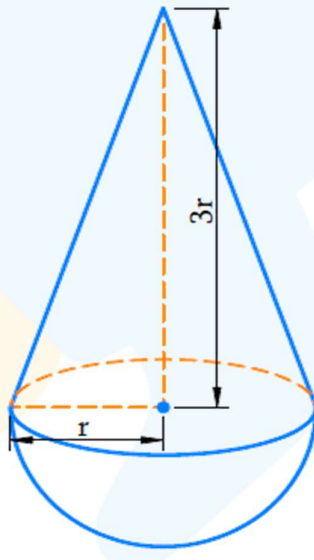
$$\frac{C_A}{C_B} = \frac{10.8}{6} = 1.8$$

$$\Rightarrow C_A = 1.8C_B = 1.8(4000) = 7200$$

A's paint job will cost INR 7200.

S23. Self-exercise. Show that a *complete volume fit* is possible. The answer will be 192.

S24. The following figure shows the structure of the toy:



The total volume of one single toy is

$$\begin{aligned} V_{\text{toy}} &= V_{\text{hemisphere}} + V_{\text{cone}} \\ &= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2(3r) \\ &= \frac{5}{3}\pi r^3 \end{aligned}$$

Since 12 toys were made from the original cylinder (whose radius was 6 cm and height was 15 cm), we have:

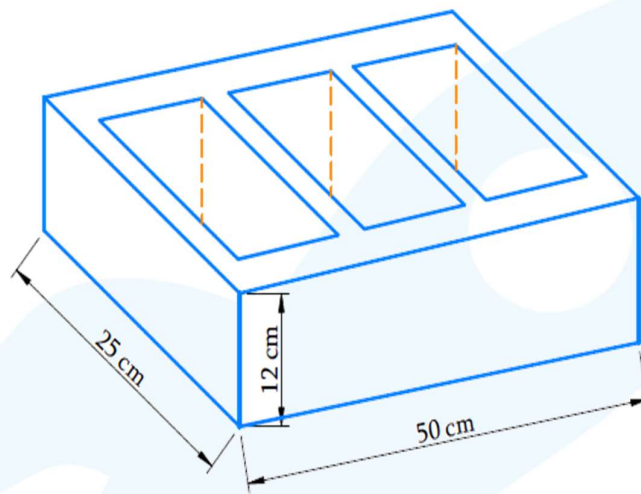
$$12 \times V_{\text{toy}} = V_{\text{cylinder}}$$

$$\Rightarrow 12 \times \frac{5}{3} \pi r^3 = \pi (6)^2 (15)$$

$$\Rightarrow r^3 = 3 \Rightarrow r = 3$$

Thus, the radius of the hemisphere is 3 cm, and the total height of the toy is $r + 3r$, or 12 cm.

S25. The following figure shows the geometrical structure of the container:



If there were no partitions, the box could have held a volume of water equal to

$$(50 - 6) \times (25 - 6) \times (12 - 3) = 7524 \text{ cm}^3$$

Visualize the partition in your mind: each partition is a cuboid with dimensions $19 \text{ cm} \times 9 \text{ cm} \times 5 \text{ cm}$. Thus, the volume inside the box, taken up by the partitions, is

$$2 \times 19 \times 9 \times 5 = 1710 \text{ cm}^3$$

Thus, the actual volume which the container can hold is

$$7524 - 1710 = 5814 \text{ cm}^3$$

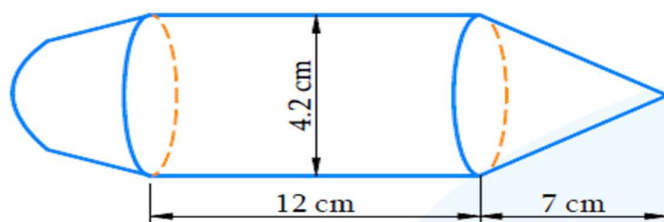
S26. The volume of air in the room is

$$\pi r^2 h \approx (3.14) \times (10)^2 \times 6 = 1884 \text{ m}^3$$

Each prisoner consumes 0.5 m^3 of air per hour (utilizes the useable oxygen). 10 prisoners will consume 5 m^3 of air in an hour. Thus, the number of hours the useable oxygen will last is

$$\frac{1884}{5} \approx 377 \text{ hours} \approx 15.7 \text{ days}$$

S27. The following figure shows the structure of the toy (the figure is not to scale):

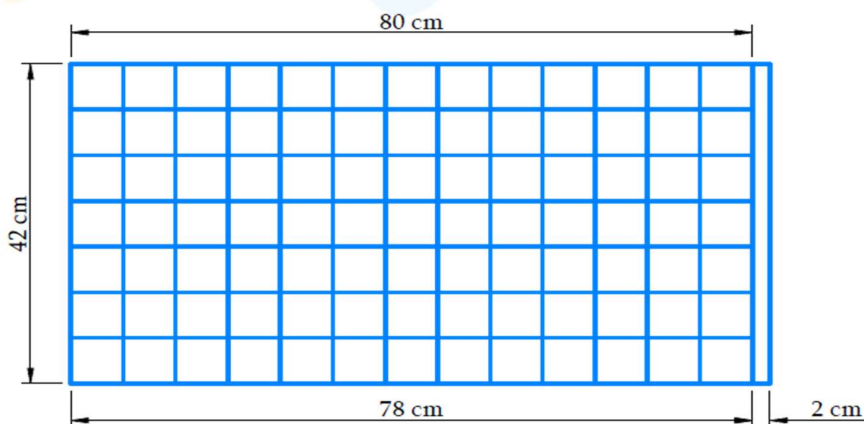


The total volume can be calculated as follows:

$$\begin{aligned} V_{\text{toy}} &= V_{\text{hemisphere}} + V_{\text{cylinder}} + V_{\text{cone}} \\ &= \frac{2}{3} \pi (2.1)^3 + \pi (2.1)^2 (12) \\ &\quad + \frac{1}{3} \pi (2.1)^2 (7) \\ &= \pi (2.1)^2 \left\{ \frac{2}{3} (2.1) + 12 + \frac{7}{3} \right\} \\ &\approx 218 \text{ cm}^3 \end{aligned}$$

S28. (a) A complete volume fit is possible – it is left to you as an exercise to figure how. The answer is 280.

(b) Now, a complete volume fit is not possible. The following figure shows a horizontal cross-section of the packing scheme:



The number of boxes which can be fit in the carton under such a packing scheme is $13 \times 7 \times 3 = 273$. The remaining volume in the carton will be $42 \times 2 \times 30 = 2520 \text{ cm}^3$ or 2.52 L.

S29. The cross-sectional area of the pipe is

$$\pi(20 \text{ cm})^2 \approx 1256.6 \text{ cm}^2$$

Thus, the volume of water coming out of the pipe's mouth per second is

$$\begin{aligned} & (1256.6 \text{ cm}^2) \times (25 \text{ cm / sec}) \\ &= 31415 \text{ cm}^3 \approx 31.4 \text{ L} \end{aligned}$$

S30. Let the minimum flow rate of the water through the pipe be $f \text{ cm / sec}$. We have (20 minutes is 1200 seconds):

$$\begin{aligned} & (f \text{ cm / sec}) \times (400 \text{ cm}^2) \times (1200 \text{ sec}) \\ &= \frac{2}{3} \pi (300 \text{ cm})^3 \\ \Rightarrow f &\approx 118 \end{aligned}$$

Thus, water must flow through the pipe at a minimum rate of about 118 cm / sec.

S31. The amount of water coming out the pipe per second is

$$\begin{aligned} & (12 \text{ cm}) \times (20 \text{ cm}) \times (50 \text{ cm / sec}) \\ &= 12000 \text{ cm}^3 = 12 \text{ L} \end{aligned}$$

Since water is also leaking at a rate of 2 L / sec, the net inflow of water into the tank is 10 L / sec. Now, the volume of the tank is

$$\begin{aligned} V &= \frac{1}{3} \pi (200 \text{ cm})^2 (100 \text{ cm}) \\ &\approx 4187 \text{ L} \end{aligned}$$

Thus, the time taken to fill the tank will be

$$\frac{4187 \text{ L}}{10 \text{ L / sec}} = 418.7 \text{ sec} \approx 7 \text{ min}$$

S32. To calculate the volume of the gold, we can multiply the surface area of the relic by the thickness of the gold layer. The surface area of the relic is

$$4\pi(200\text{ cm})^2 \approx 502400\text{ cm}^2$$

Thus, the volume of gold in the gold layer is

$$(502400\text{ cm}^2) \times (0.2\text{ cm}) = 100480\text{ cm}^3$$

The mass of all this gold is

$$\begin{aligned} & (100480\text{ cm}^3) \times (20\text{ g / cm}^3) \\ &= 2009600\text{ g} \end{aligned}$$

Thus, the cost of this gold will be

$$\begin{aligned} & \frac{(2009600\text{ g})}{10} \times (\text{INR } 25,000) \\ &= \text{INR } 502.4\text{ crores} \end{aligned}$$

Clearly, an extremely valuable loot!

